

A Transdisciplinary Approach to Reducing Transportation Costs and Enhancing Production Planning in Industrial Chemical Services

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Abstract. This paper describes a project conducted in a company that provides chemical water treatment services for industrial cooling systems across Mexico. The project addresses the challenge of rising transportation costs and inefficient production planning in a dynamic service environment. Key stakeholders, including the company's General Director and Plant and Logistics Manager, identified transportation expenses as the fastest-growing operational cost, with increases more than double the inflation rate in Mexico over the past five years. Compounding this issue, the company currently operates with reactive production planning, relying on monthly client estimates and lacking effective control over shipment management, resulting in demand imbalances. The project utilizes a transdisciplinary approach, combining strategic planning and statistical analysis to determine the priorities of the company and establish a framework for demand planning and transportation management. The proposed approach is founded in methodologies such as Hoshin Kanri, project management, time series analysis and control charts. The main results of the project include: (1) a cost-reduction strategy for transportation, achieving a 20-25% decrease in associated expenses; and (2) a set of rules for proactive production planning and shipment management, which enable earlier procurement and help identify demand fluctuations. By integrating quantitative analysis with business management practices, this work proposes a sustainable solution to logistical challenges, aligning production planning with demand patterns and reducing costs. This research aligns with the transdisciplinary engineering methodology by bridging statistical analysis with operational strategies to create value across business and logistics processes.

Keywords. Strategic planning, Hoshin Kanri, Project management, Transportation, Transdisciplinary engineering.

Introduction

Organizational competitiveness is heavily influenced by the efficiency of a company's operations and logistics. Small and medium-sized enterprises (SMEs) are continually striving to achieve this efficiency; however, their limited level of professionalization and process design often pose significant barriers. This paper presents a case study of an SME

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in the chemical industry that has experienced a significant increase in transportation costs in recent years, threatening its competitive position.

The company under study is a family-owned business founded in October 2000. It specializes in providing chemical water treatment services, primarily for industrial cooling systems. These include open systems, such as cooling towers, and closed systems, such as compressors and air-cooled heat exchangers. The company serves more than 30 clients across Mexico, most of which are involved in manufacturing PET plastic bottles for the beverage industry. The company manufactures and distributes the necessary chemical products from its production plant and distribution centre located in Toluca, Estado de México (in the central region of Mexico). Clients are grouped into three regions: North (5 cities), Central (5 cities), and South (2 cities).

The company offers an “in-house” service model, delivering the required chemical products and tools directly to the client’s facility. A service technician from the company visits the client’s site between one and six times a week, depending on the specific characteristics and equipment of each facility.

The company’s current operations are structured as follows. At the end of each month, technicians from various cities submit inventory information for each client and request sufficient products to meet the estimated demand for the following month, ensuring a small safety stock. Once this information is received at the plant, the requested products are shipped during the first ten days of the month. The remaining days are allocated to producing the required inventory for the next month. Production planning involves raising the plant’s inventory to the maximum estimated monthly consumption level, along with an additional safety stock buffer.

In recent years, the company has stopped expanding its client base due to reaching its maximum on-site service capacity, even though the production plant still has significant unused capacity. The number of clients it can serve is limited by the director’s ability to manage over 30 employees remotely and ensure that work is performed correctly. According to the director, the company’s primary competitive advantage lies in the quality of its service. On the other hand, the transportation cost has increased 69% in the last three years.

The lack of processes carefully designed, and the increasing transportation cost represent a threat to the company. It is essential to conduct an in-depth analysis of its operational challenges and define the strategic directives it should focus on in the coming years. To address these challenges, this paper presents a project aimed at defining the organization’s direction for the coming years while addressing demand management to mitigate rising transportation costs. To achieve this, the project incorporates methods and techniques such as Hoshin Kanri, Project Management, and Statistical Analysis to develop a strategic plan and translate it into operational directives.

1. Methods

This section introduces the methods used in the project. Although the project is not research-based, it is founded in well-known methodologies about strategic planning, project management and statistical analysis. These methods were integrated to design a comprehensive solution to multiple challenges faced by the company. Next, we briefly describe the methods used in the approach.

1.1. Project management

To have a clear roadmap, the project management followed the guidelines of the Project Management Institute (PMI), which established the methods, procedures and tools to manage a project. These guidelines are part of the handbook “A Guide to the Project Management Body of Knowledge PMBOK®” [1].

The PMI methodology consists of five project stages (initiation, planning, execution, monitoring and controlling, closing, as depicted in Figure 1), 10 knowledge areas, and 49 processes.

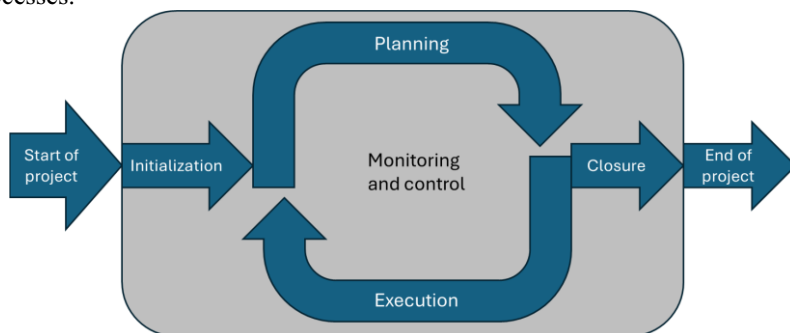


Figure 1. Project stages in project management [1].

During the initialization stage, a *Project Charter* is developed and completed during the planning stage. This is considered the project's official document, and it outlines the scope, milestones or expected events, dates, associated operating costs, budget, key resources, stakeholders, associated risks, project phases, among others. The signature of the *Project Charter* by the sponsor is considered the formal start of the project [1].

In the planning stage, the WBS (Work Breakdown Structure) is developed. As its name indicates, this document represents the decomposition or subdivision of the project into phases, deliverables, work packages, or smaller projects that clarify the work to be done and facilitate its management. During the development of the WBS, the cost and duration of each associated subdivision must be estimated. This serves as a basis for establishing the project's resources and activities, as well as for the proper monitoring and control of changes to the stipulated scope [1].

1.2. Strategic planning

Strategic planning is embodied in the decision-making process, allowing companies to replace their reactive approach with a proactive one that enables them to anticipate changes, mitigate risks, develop their strengths, and achieve success [2].

The first steps in strategic planning are the definition of the company's philosophy, vision, and mission [3]. The definition of a company's philosophy is important both for strategy formulation and for defining the mission. A strong business mission should include the organization's philosophy. The business mission should answer the question: What is our business?, as this defines the purpose that distinguishes the organization from others. It should also state whom the organization aims to serve and what products or services it offers [2].

The vision is the statement of what the company aims to achieve in the long term. It should be clear and reflect the type of business the company is involved in. A clear vision answers the question, "What do we want to become?" [2].

Once that the philosophy, mission and vision are established, the strategic guidelines that include long-term, and short-term objectives are determined. In this case, we used the Hoshin Kanri methodology to define these guidelines. From the Japanese “Hoshin” (direction) and “Kanri” (management, administration, control), the term can be interpreted as “management by policy” or “management by direction.” [4]. The importance of the Hoshin Kanri methodology lies in its versatility—it can be used as a strategic planning method, a project management tool, or a cross-functional management approach.

Hoshin Kanri is based on the concept of putting principles into practice by translating strategic directives into key projects or activities. It also serves as an effective planning tool for aligning the different levels of an organization toward strategic goals [4].

1.3. Statistical analysis

The analysis and actions designed to improve and control the current processes are based on evidence. Therefore, statistical analysis is required to identify trends and factors that affect the current performance of the process and determine actions that might improve them. The main methods used in this case are time series analysis and control charts.

Time series refers to observations of a variable collected over time. Given that data are collected through successive periods, time series observations are related to one another (autocorrelated), which allows for the identification of variability patterns that can be used to forecast future values. In business, time series forecasting helps to understand past data and make plans that enable proactive responses to the future [5].

On the other hand, control charts are tools that help to monitor process and identify assignable causes of variation. That is to say, the control charts help to identify abnormal situations in a process by using a continuous monitoring approach.

1.4. Proposed approach

In this project, the methods described above were combined into a comprehensive approach to analyse the challenges faced by the companies and design solutions that would help the company to keep competitiveness.

First, a strategic planning exercise was executed to define the mission, vision, and objectives of the company, as well as to identify the essential processes, the context (internal and external) surrounding the company, and the strengths, weaknesses, opportunities and threats. This exercise involved multiple stakeholders of the company, and it was supported by the methodologies Hoshin Kanri and Soft Systems Methodology.

The work on the strategic planning revealed two issues in the company: 1) the increasing transportation cost from the manufacturing plant to the location of the clients, and 2) a reactive production and inventory management. The next step consisted of analysing the demand, production and inventory policy, and distribution plan to propose new distribution schemes and production planning. For this, statistical analysis methods were used, including time series analysis to study and forecast the demand, and control charts to define a production and inventory policy. In addition, the distribution network was studied using cost analysis. An alternative distribution network was proposed to reduce the transportation cost while keeping the same service level. All these methods were combined in a framework that was managed through the guidelines of project management. Figure 3 shows a representation of the proposed approach that includes the

methods and the outcomes of each stage. Next, we briefly describe the findings throughout the project.

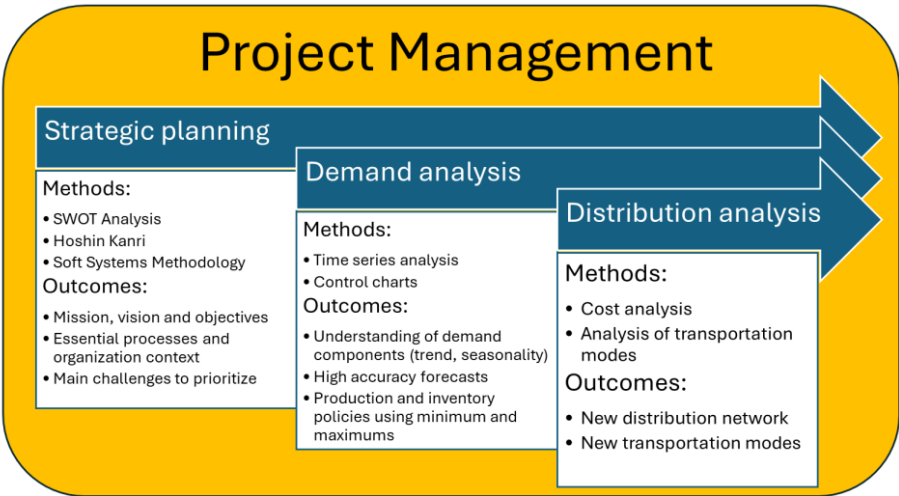


Figure 2. Methods and outcomes of the proposed approach.

2. Project Development and Outcomes

In this chapter, we summarize the process followed to develop a strategic plan for the company under study, based on the methods and tools described in the previous chapter. In addition, we describe some of the outcomes obtained at different stages of the project.

2.1. Project start

The project began with the development of the *Project Charter* and the identification of stakeholders and interest groups. All individuals or organizations impacted by the project were identified, to later document their interest and the level of involvement required for the project's success. Table 1 shows the identified stakeholders according to their position (internal or external) and their impact (on the process or the project outcome). A total of 7 stakeholders were identified.

Table 1. Identified stakeholders.

Impact	Internal Stakeholders	External Stakeholders
Impacted during the project execution	Project manager	
	Company director	
	Production plant director	
Impacted by the project outcomes	Service staff	Courier company
	Service technician	Customers

Once all stakeholder groups were fully identified, their levels of power, interest, influence, and required involvement throughout the project were assessed, as these factors determined the actions, considerations, and communications established during the project's execution.

In the project planning stage, a SWOT analysis was conducted to identify the organization’s main Strengths, Opportunities, Weaknesses, and Threats [6]. The SWOT matrix provided insight into the company’s current situation and served as the starting point for identifying the key requirements to be addressed as part of the strategic planning process. Table 2 shows the SWOT matrix with the prioritization of the main Strengths, Opportunities, Weaknesses, and Threats that the company should focus on as part of its strategic planning.

Table 2. SWOT matrix prioritization.

Strengths	Opportunities
Strong service quality and company reputation as a problem solver.	Growth in Mexico's industrial and service sectors.
Focus on outcome-based service (corrosion, scaling, bacteria prevention).	High water salinity in northern Mexico increases demand for specialized water treatment (anti-corrosive/scaling).
High technical knowledge in water treatment.	
Presence in key industrial cities (Guadalajara, Mexico City, Monterrey, Tijuana, Querétaro).	
Plant has capacity to double production	
Weaknesses	Threats
High service cost (generally perceived as expensive).	Increasing employee turnover in recent years.
Poor internal and external information management.	Company reaching full-service capacity, risking a perceived decline in performance.
No training plan for technicians and service leads.	Increasing transportation costs threaten operational efficiency.
Undefined internal processes.	
Lack of continuous customer satisfaction monitoring.	
Lack of awareness of max/min monthly client shipments.	

The SWOT matrix revealed that the overall objective of the project should be to lay the foundation for the company to continue achieving sustainable growth in the coming years by expanding its client base with operational efficiency, without compromising service quality. The whole set of specific objectives included the following:

1. Redefine the company’s mission and vision in line with its current objectives and expectations.
2. Redefine the company’s values according to its current objectives and expectations.
3. Redefine the company’s quality policy in accordance with its current objectives and expectations.
4. Define the systemic mapping of the company and its core processes based on current characteristics and expected goals.
5. Define the company’s main strategic guidelines aligned with the overall objective.
6. Analyse product demand behaviour by customer/region, identifying trends and seasonality.
7. Establish control limits for monthly shipments.
8. Analyse transportation cost behaviour.
9. Identify transportation alternatives to improve operational cost efficiency.

After defining the objectives, the work breakdown structure (WBS) was created as a starting point for planning the project's milestones and activities. As its name suggests, it represents the breakdown of the project into subdivisions of phases, work packages, and activities, which facilitated management and provided clarity for planning. Once the activities were defined, the cost, duration, and associated resources for each of them were estimated.

2.2. Strategic planning phase

The strategic planning analysis initiated using *Soft Systems Methodology* (SSM), to define the systemic mapping, core processes, and company philosophy, due to its systemic and holistic approach to managing complex problems [7].

The mission was formulated using the “essential definition” process from SSM, considering the company as the fundamental system to be analysed. The essential definition is a concise statement that describes what a system is in its most fundamental form and serves as the foundation for establishing an intentional human activity model. A well-formulated essential definition should include the six elements of the CATWOE acronym: Customers, Actors, Transformation, Weltanschauung (worldview), Owners, and Environment (constraints) [8]. This analysis is presented in Figure 3.

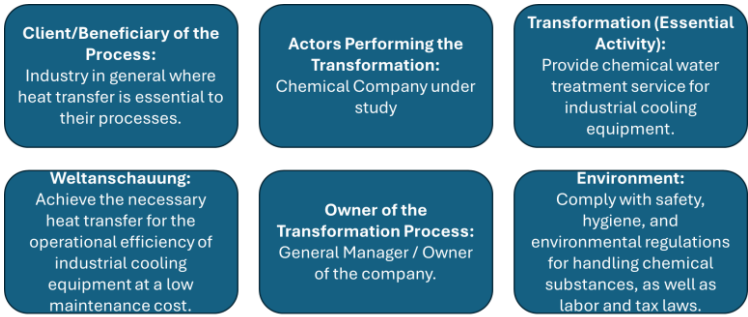


Figure 3. CATWOE analysis.

Based on the CATWOE, the mission formulated for the company is:

“To provide chemical water treatment services for industrial cooling equipment, ensuring service quality by achieving the heat transfer necessary for the efficient operation of equipment at a low maintenance cost. The company accomplishes this through the inhibition of corrosion, scaling, sludge fouling, and bacteriological growth, while also promoting proper water use and compliance with safety, hygiene, and environmental standards in the handling of chemical substances, as well as with labor and tax laws.”

While the vision was defined as:

“To be a company recognized internationally for its excellence in water treatment products and its quality service focused on achieving operational efficiency in industrial cooling equipment. We seek to work with leading companies in each industry, acting as operational partners in their success within the industrial sector, with the goal of achieving operational excellence for their equipment.”

The systemic mapping of the company allowed us to identify the essential process of the company, as shown in Figure 4. Then, the CATWOE analysis was performed for each process to understand how they operate.



Figure 4. Essential processes of the company.

As the second part of the strategic planning process, the Hoshin Kanri methodology was used to define the direction of the organization, considering the strategies identified in the SWOT matrix, along with the previously defined philosophy, vision, mission, and core processes. The strategic guidelines are considered the key policies for the organization. Four strategic guidelines were identified: quality of service, customer satisfaction, sales, finances. Figure 5 shows the strategic objectives for each of these guidelines.

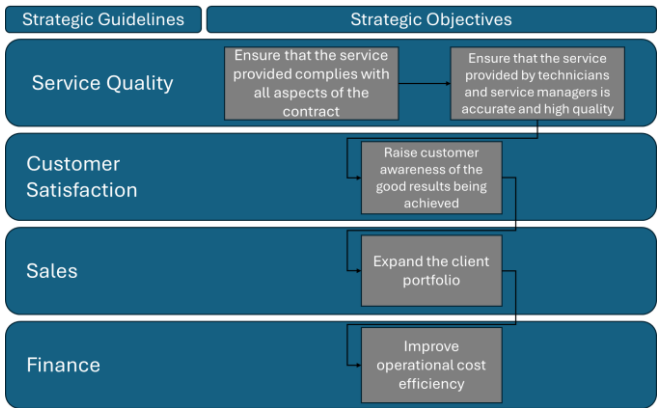


Figure 5. Strategic guidelines and strategic objectives.

These strategic guidelines are aligned to two processes: demand planning and product transportation. Next part consists of analyzing these processes.

2.3. Demand planning and product transportation analysis

The current distribution network is shown in Figure 6a. This figure shows that the current approach is direct shipment from the production plant to each city where customers are located. All products are shipped to customers through parcel services using consolidated transportation. Based on the demand and location analysis of the customers, four routes were identified to cover 91% of the total demand. A new transportation network is proposed to send the products from the production plant to distribution nodes on four routes that would distribute the products to other customers in their zone. The proposed network is shown in Figure 6b.

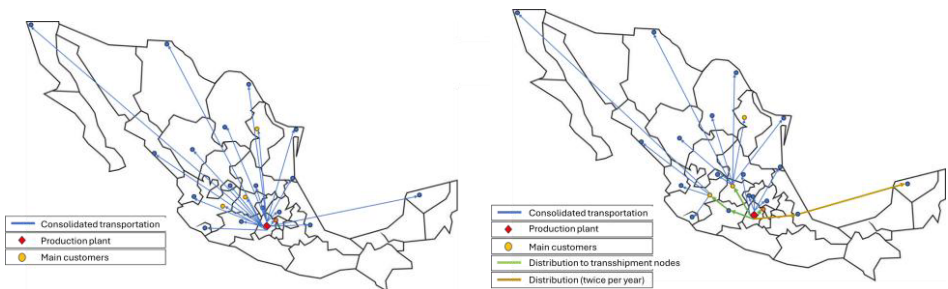


Figure 6. Distribution network. a: current (left), b: proposed (right).

The demand planning for each route was established using time series analysis. For instance, consider the Guadalajara route that represents 17% of the total sales of the company. This route would have the transshipment node and would distribute the product to customers located in eight cities in the northwestern zone of Mexico. Analysing the demand for the customers of this graph, a stationary time series with yearly seasonality was identified. January and June show the highest peaks in the prediction model, with +30% and +7% higher demand (2,473 kg and 2,022 kg) compared to the average monthly sales (1,893 kg), while September and December have the lowest peaks in the prediction model, with -12% and -7% lower demand (1,672 kg and 1,741 kg) compared to the average monthly sales. A time series decomposition model can fit the demand pattern with a mean absolute percentage error of 9.2%, which is a reasonable value for the company. Figure 7 shows the time series analysis of the total demand per month between 2019 and 2023, with the forecasted demands for each month of 2024.

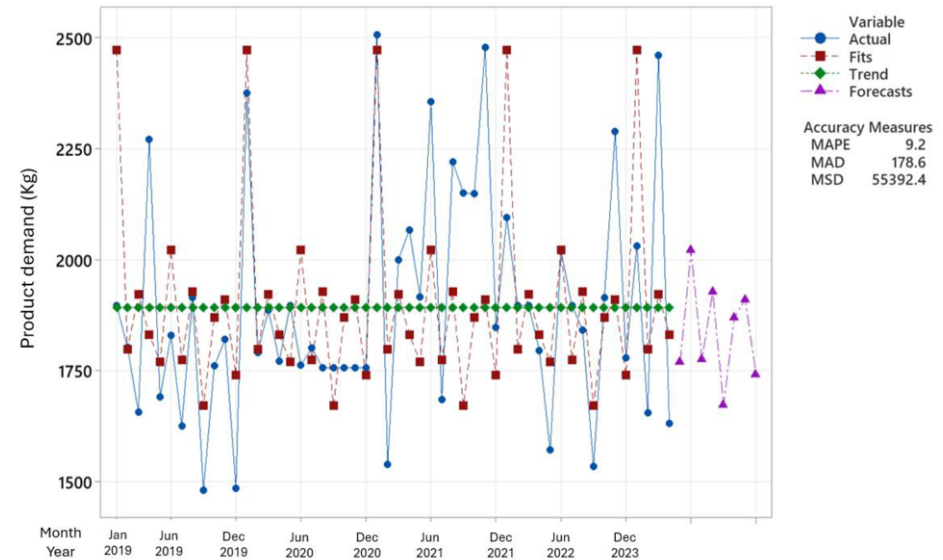


Figure 7. Time series decomposition of total demand (output from Minitab).

Given the stability of the demand, control charts for sample averages were used to monitor the demand and identify abnormal variation in the demand.

Once a similar analysis has been done for all the routes, a financial analysis was performed to estimate the cost of the new distribution network. For that, several options were considered, including outsourcing the transportation (exclusive and consolidated services), and leasing trucks. The results show that a hybrid strategy would reduce the transportation cost per year by 20%-25%.

2.4. Project closure

The project closure consisted of completing the activities and deliverables outlined in the project planning phase. During this stage, the following activities were concluded in collaboration with the project team.

First, it was ensured, together with the stakeholders and project advisors, that the stated objectives were achieved through sessions in which the acceptance and approval of the analyses were confirmed.

Second, together with the general director, it was concluded that the implementation of the routes and transportation alternatives is a viable option for the company. It was agreed that the next phase of the project could begin implementation in January 2025.

Third, at the end of the project, a retrospective session was held with the general director and plant manager. Among the lessons learned, it was agreed to maintain a more detailed record of monthly data regarding the number of products shipped, the quantity stored at client locations, and transportation costs, as this information will be essential for future analyses like those carried out during the project.

Additionally, it was agreed that an annual review will be conducted of product demand, transportation routes, and freight costs, with the goal of continuously improving operational efficiency and keeping production, storage, and shipping management rules updated in line with changing demand.

3. Conclusions

This paper presented an interdisciplinary approach to combine elements from project management, strategic planning and statistical analysis to assess a chemical company in the formalization of its processes. The main reason for this project was a significant increase in transportation costs, and a limit on the expansion of the company.

The methodology used a systemic and holistic approach that aligned the main stakeholders since the identification of the philosophy, mission, vision and essential processes of the company, until the analysis of demand and transportation to generate and evaluate new production and distribution strategies.

The diagnostics and recommendations generated through this project would help the company to overcome their current challenges and plan a sustainable future.

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